

Fall 2026

Lab 2: HRI Research Methods

Due: October 1, 2026 at 11:00PM

Assignment Type: Group (Up to 4 members)

Submit PDF on Gradescope; Entry code: 2EK5JN

Total Points: 34 points

0. Overview and Learning Goals

HRI research methods are critical for understanding the influence of robotic technology on people and society. Strong experimental design, execution, and reflection on results enable HRI researchers and developers to validate the benefits of robotic innovations.

Research methods let HRI researchers move from an interesting robot behavior to a solid claim about robotics. In this lab, your team will design and run a small, controlled pilot study using the Reachy Mini. You will manipulate one robot-behavior factor across two conditions, collect mixed-methods data, and report what the evidence does and does not support. Through this process, you will gain firsthand experience with the complexities involved in designing and conducting an HRI user study. The **learning outcomes** of Lab 2 are:

- Run teleoperation on physical robot
- Create and run an app on the physical robot
- Design and conduct a user study to answer 2-3 research questions
- Analyze and reflect on key findings
- Write a scientific report to summarize the study and findings

1. Run Teleoperation on the Physical Robot

1.1 Connect to RedRover WiFi

Connect to the RedRover WiFi [here](#) (Click on the ‘Register an IoT Device on RedRover’ button on the left-hand side of the website). Register the physical Reachy Mini using the device identifier. Your laptop and the Wireless Reachy Mini must be connected to the same approved network for local discovery and control. Get started by following the ‘**Lab 2 - Reachy Mini Access and Basic Testing Guide.pdf**’ on Canvas→Home→Course Logistics for Reachy Mini setup on your virtual machine.

1.2 Connect and check system

Power on the assigned Reachy Mini and wait for startup. Open Reachy Mini Control or <http://reachy-mini.local:8000/>. Confirm battery level, motor state/temperature, and system status.

The Wireless robot already runs its daemon; do not start a second daemon on the robot.

1.3 Run *teleoperation* on physical robot

- Run the [Reachy Mini Telepresence](#) application on the physical robot.
- Clear at least 30 cm around the robot and assign one teammate to the stop control.
- Use teleoperation to make one small head movement and one small antenna movement.
- Return the robot to neutral and stop manual control before launching the custom app.
- **There are two options for running the Teleoperation app, either 1) from the Reachy Mini Control App or 2) download the code and run it locally onboard the robot from the terminal.**

1.4 Deploy the final simulated candidate

- For the first physical run, reduce motion amplitudes to no more than half of the final simulated values.
- Run one cycle while the operator remains at the stop control. Check clearance, smoothness, timing, and neutral recovery.
- If the first run is safe, make one documented parameter adjustment and run one final recorded cycle.

Deliverables (3 points):

- A short clip showing the safe final app cycle on the physical robot, including neutral recovery.
- A screenshot or log excerpt showing the same stage markers used during simulation.
- A brief sim-to-real comparison naming at least two observed differences and any final parameter adjustment.

2. Create code for a hello world application on the robot.

Make a python program called 'hello_world.py,' that moves to robot's antennas from a starting position to a fixed position, and store the following code in the file:

```
from reachy_mini import ReachyMini

with ReachyMini() as mini:
    mini.enable_motors()

    print("Connected, moving antennas...")

    mini.goto_target(
        antennas=[0.3, -0.3],
        duration=1.0
    )

    mini.goto_target(
        antennas=[-0.3, 0.3],
```

```

        duration=1.0
    )

    mini.goto_target(
        antennas=[0.0, 0.0],
        duration=1.0
    )

    print("Done")
...

```

Make one change to the code, including modifying the robot's gaze, head, base orientation, movements based on camera-based inputs (move based on sensor detection of hands, faces etc), or audio inputs (e.g., move based on natural language inputs). Other options to explore are [here](#).

Deliverables (3 points):

- A short clip showing a demonstration of the robot in action.
- A screenshot or log excerpt showing the same stage markers used during simulation.
- A brief description of changes made to the code, expected outcomes and actual outcomes. Note that perfect results are not necessary. Instead, communicate what you learned from testing the robot.

3. Create and run custom app on the physical robot

3.1 App Selection

Select capabilities from at least two Reachy Mini apps or app components from the Reachy Mini store / GitHub and integrate them into one custom app. Your selections must support an HRI topic that your team could continue for the final project, such as expressive robot behavior, social signaling, turn-taking, attention, accessibility, education, or telepresence

- State the potential final-project topic or problem area and explain how the selected app capability contributes to that direction.
- Choose a combination that can support a focused human-robot interaction

3.2 Integration and Experimental Manipulation

Combine the selected capabilities into one coherent interaction. Create two reproducible conditions that differ in one robot-behavior factor, such as response latency, movement speed, motion amplitude, or repetitions.

- Expose the selected factor as one clearly named parameter with documented units as: Condition A and Condition B.
- Keep the sequence, operator behavior, environment, and all other robot settings constant.

3.3 Testing, Logging, and Reproducibility

- Test both conditions in the simulation first and then on the physical robot. Confirm that

- each trial begins and ends in the same pose.
- Log the condition, parameter value, start time, end time, completion status, and any robot or operator error for each trial.
- Document the setup instructions, launch command, and stop procedure.

Deliverables (4 points):

- A working two-condition Reachy Mini app with one named manipulation parameter.
- A brief integration map identifying the source apps, the capabilities reused or modified, the interaction goal, and its connection to a potential final-project topic.
- Reproducible setup and launch instructions, including the exact values used
- A short robot-only demonstration showing one complete cycle of each condition.

4. Design 1-2 HRI research questions

4.1 Frame the HRI Problem

Define the HRI problem, explain its significance, and ground the proposed study in relevant theory and prior research.

4.2 Define and Operationalize the Constructs

Identify the central constructs and define how each will be manipulated or measured. Specify the independent variable, outcomes, manipulation check, controls, potential confounds, measurement scales, and evidence of construct validity.

4.3 Formulate the Research Questions or Hypotheses

Develop one or two research questions or hypotheses that follow directly from your framework. Clearly distinguish exploratory research questions from confirmatory hypotheses. Hypotheses must state an expected direction and be supported by theory or prior evidence.

Example research question: How does faster versus slower head movement affect participants' ratings of Reachy Mini's perceived animacy during a greeting interaction?

Example hypothesis: Participants will report higher **Human–Robot Interaction Evaluation Scale (HRIES)** animacy scores in the faster head-movement condition than in the slower condition.

Deliverables (3 points):

- A concise, research-grounded problem statement.
- A construct map linking the claim, manipulation, measures, and expected evidence.
- Operational definitions of the variables, controls, and potential confounds.
- One or two clearly identified research questions or hypotheses.

5. Design and Conduct an HRI User Study

Design a controlled within-subjects pilot in which each participant experiences both robot conditions. Recruit at least four adult volunteers who are not members of your team. Standardize the protocol, counterbalance condition order, and document every deviation to support interpretable within-participant comparisons.

5.1 Study Design / Conditions

Use two conditions that differ only in the selected robot behavior. Counterbalance order to mitigate sequence and carryover effects. For four participants, use AB, BA, AB, BA and assign the sequence before each session begins.

Element	Condition A	Condition B	Held constant
Primary factor	Baseline value and units	Comparison value and units	Manipulated factor and permitted range
Robot behavior	Behavior Sequence	Same Sequence	Start post, timing, and return
Interaction	Participant Task	Same Task	Prompt, setting, and operator

5.2 Target Population, Participant demographics, Robot familiarity

- Define the target population and justify the use of a convenience sample.
- Recruit at least four adult participants who are not members of your team.
- Collect only relevant demographic information and prior familiarity with robots.
- Assign each participant a de-identified ID, such as P01 or P02.
- Explain how the sample size and recruitment method limit the transferability of the findings.

5.3 Study Task

Design a short, repeatable task that makes the manipulation salient without revealing the hypothesis. The task should elicit meaningful engagement with the robot, require no specialized expertise, and take approximately three to five minutes per condition.

- Keep instructions, prompts, interaction sequence, timing rules, and completion criteria constant across conditions.
- Operationally define completed, interrupted, failed, and repeated trials before data collection.
- Minimize demand characteristics and experimenter expectancy by using neutral instructions and standardized facilitation.

5.4 Study Setting

Standardize the room layout, robot and participant positions, lighting, ambient noise, display, instructions, and operator location. Record disruptions or environmental variation to ensure that changes by external factors are documented.

5.5 Study Procedure

Develop a facilitator script and execute the following standardized sequence for every participant:

1. Prepare the room and robot; verify both conditions, logging, and the robot control.
2. Explain the activity, get agreement to participate, and answer questions without revealing the hypothesis or expected result.
3. Assign the participant ID and preassigned AB or BA order.
4. Read the task instructions and place the participant at the marked location.
5. Run the first condition and collect the objective and post-condition measures.
6. Return Reachy Mini to neutral, then run the second condition using the same procedure.
7. Collect the study data then debrief the participant and explain the study purpose.
8. Examine the session record for completeness and document missing data, interruptions, robot faults, or protocol deviations without imputing unobserved values.

5.6 Subjective, Objective, and Qualitative Measures

Standardized questionnaires support reliable and comparable HRI evaluation. [This Database](#) provides validated measures for a range of constructs.

For this study, use the **Human–Robot Interaction Evaluation Scale (HRIES)** to measure participants' perceptions of the robot across four dimensions: sociability, animacy, agency, and disturbance. [This article](#) demonstrates its use in an HRI study, and [the supplementary materials](#) provide the questionnaire, response scale, and administration procedure.

Collect three forms of evidence after each condition:

- **Subjective quantitative measure:** Administer the 16-item HRIES using its published 7-point scale. Identify one HRIES dimension as the primary outcome and calculate each dimension separately. Include one additional item confirming that participants perceived the intended behavioral difference.
- **Objective behavioral measure:** Record one observable outcome, such as response time, task duration, number of prompts, identification accuracy, or task success.
- **Qualitative measure:** Ask one neutral, open-ended question about the participant's interpretation and experience of the interaction.

Deliverables (7 points):

- A completed conditions table identifying the manipulated robot behavior and all controlled elements.
- The HRIES questionnaire, one manipulation-check item, one objective behavioral

- measure, and one neutral open-ended question.
- A description of the target population, sampling rationale, relevant demographics, robot familiarity, and counterbalanced condition assignments for at least four participants.
- A repeatable study task with standardized instructions, completion criteria, and documented environmental controls.
- A facilitator script covering setup, participation agreement, condition order, data collection, debriefing, and protocol deviations.
- The HRIES questionnaire, one objective behavioral measure, and one neutral open-ended question.
- De-identified HRIES responses, objective and qualitative data, condition order, and a record of interruptions, missing data, or robot faults.

6. Prepare the Research Data

6.1 Organize the Data

Preserve the original responses as a read-only raw dataset and create a separate analysis-ready dataset. Each record should include the participant ID, condition, condition order, HRIES responses, objective measure, qualitative response, and any protocol deviation. Do not include names or contact information.

6.2 Score and Document the Measures

- Create a data dictionary defining each variable, response scale, units, and missing-data code.
- Check participant IDs, condition labels, AB/BA assignments, and missing or invalid values.
- Calculate separate HRIES scores for sociability, animacy, agency, and disturbance by averaging the four items in each dimension.
- Do not combine the four HRIES dimensions into a single overall score.
- Document all exclusions, corrections, and missing values without replacing unobserved data.

Deliverables (2 points):

- A de-identified raw dataset and analysis-ready dataset.
- A data dictionary defining all variables, scales, units, and missing-data codes.
- Reproducible calculations for the four HRIES dimensions and the objective measure.
- A record of missing data, exclusions, corrections, and protocol deviations.

7. Analyze and Reflect on Key Findings

Analyze the data to identify patterns and evaluate the study.

7.1 Analyze the Quantitative Evidence

- Report the final sample size, condition order, missing data, exclusions, and protocol deviations.
- For each condition, summarize the four HRIES dimensions using appropriate descriptive statistics.
- Calculate within-participant differences between Conditions A and B for each relevant HRIES dimension.
- Compare the objective behavioral measure across conditions.
- Create at least one participant-level figure showing responses in both conditions with paired observations connected.
- Interpret each HRIES dimension separately and identify whether higher values represent greater sociability, animacy, agency, or disturbance.

7.2 Analyze the Qualitative Evidence

Develop two to four categories that capture recurring explanations or interpretations in the open-ended responses. Define each category and apply the categories consistently across responses. Use de-identified excerpts or paraphrases to illustrate the findings.

7.3 Interpret and Reflect

- Answer each research question using quantitative, behavioral, and qualitative evidence.
- Describe the evidence as supportive, contradictory, mixed, or insufficient.
- Identify convergence or disagreement among the three forms of evidence.
- Discuss alternative explanations and at least three threats to validity, such as sample size, convenience sampling, order effects, measurement validity, robot variability, or experimenter influence.
- Propose one improvement to the robot interaction and one follow-up study.

Deliverables (5 points):

- A results table summarizing HRIES and objective measures by condition.
- At least one participant-level paired figure.
- A qualitative coding table defining each category and summarizing the findings.
- An evidence-based response to each research question or hypothesis.
- A reflection addressing conflicting evidence, alternative explanations, threats to validity, interaction improvements, and a follow-up study.

8. Communicate the key findings

8.1 Report

Submit a polished report. The report must present a coherent argument connecting the HRI problem, research questions, study design, evidence, and conclusions.

Include the following sections:

- **Introduction:** HRI problem, relevant prior research, study motivation, research questions, and hypotheses.
- **Method:** Reachy Mini application, experimental conditions, participants, task, setting, procedure, HRIES, objective measure, qualitative prompt, and analysis approach.
- **Results:** Participant accounting, HRIES results, objective findings, qualitative findings, figures, missing data, and protocol deviations.
- **Discussion:** Answers to the research questions, interpretation of the evidence, alternative explanations, limitations, design implications, and proposed follow-up study.
- **References:** Complete citations for HRIES and all supporting literature.
- **Appendix:** Conditions table, facilitator script, survey materials, data dictionary, and other participant-facing materials.

8.2 Submission

Submit the report and all materials required to reproduce the study and analysis. Ensure that the report, dataset, figures, and repository use identical condition names, parameter values, sample counts, and results.

Deliverables (5 points):

- A final report in PDF format (2 points).
- A complete repository containing the Reachy Mini application, study materials, de-identified data, analysis files, figures, logs, and condition demonstrations (2 points).
- A README explaining how to run the application and reproduce the analysis (1 point).

9. Extra Credit (0.5 Credits)

The objective of this form is to collect feedback on the Lab 2 assignment. The goal of the teaming team is to provide an interesting, engaging, and relevant experience for students in the course. This survey instrument serves as a measurable tool to improve the course in the future.

[Complete form here.](#)